Awesome UI with .NET MAUI





David Ortinau

@davidortinau

davidortinau@noc.social

david.ortinau@microsoft.com

MODERN CLIENT

David Ortinau

Principal Product Manager

St. Louis, MO – Central Time

Help customers succeed delivering client app experiences across various screens using .NET SDKs for mobile and desktop cross-platform – aka .NET MAUI (and Xamarin).

I like to trail run and have completed multiple distances up to 50 miles.

Grandfather. Doesn't age.



SIDE IMPACT
PROTECTION
doona


AUSTRIA
AFGHANISTAN
ALBANIA

Today's List of Awesome UI

Accessibility

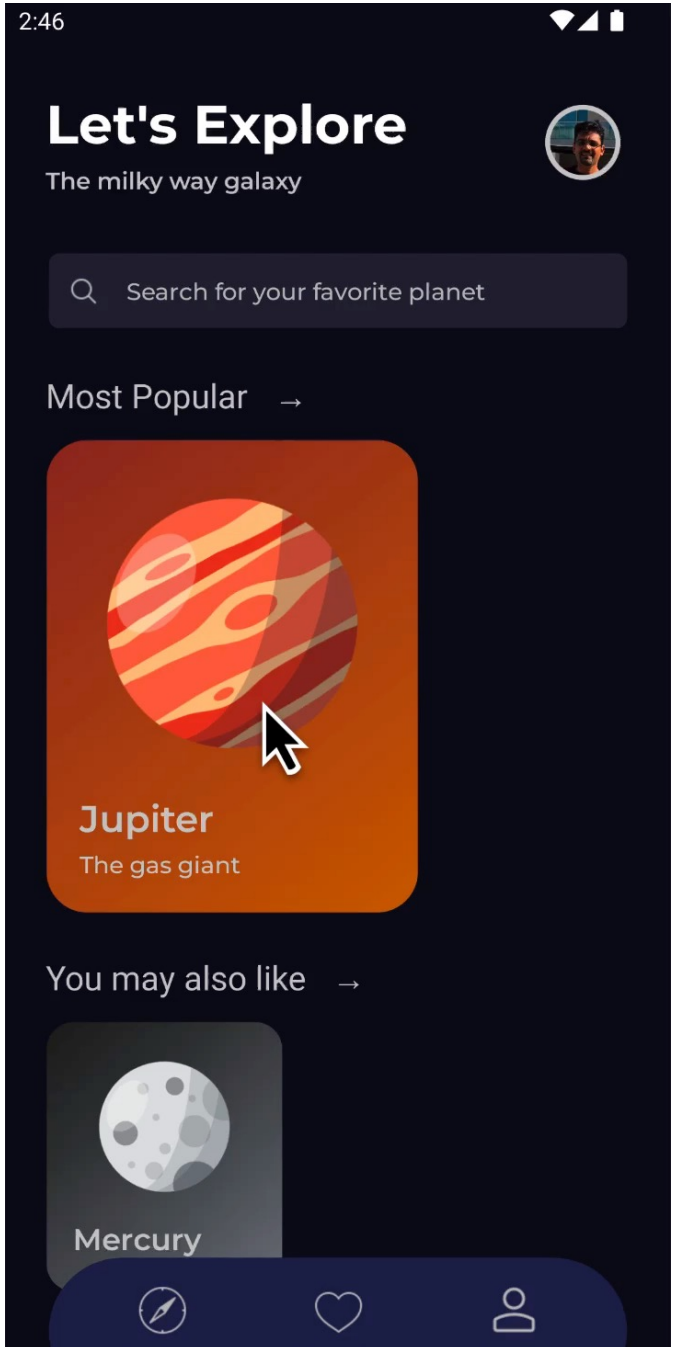
Styles and Themes

Navigation – flyout menu, tabs

Control customizations

State Container

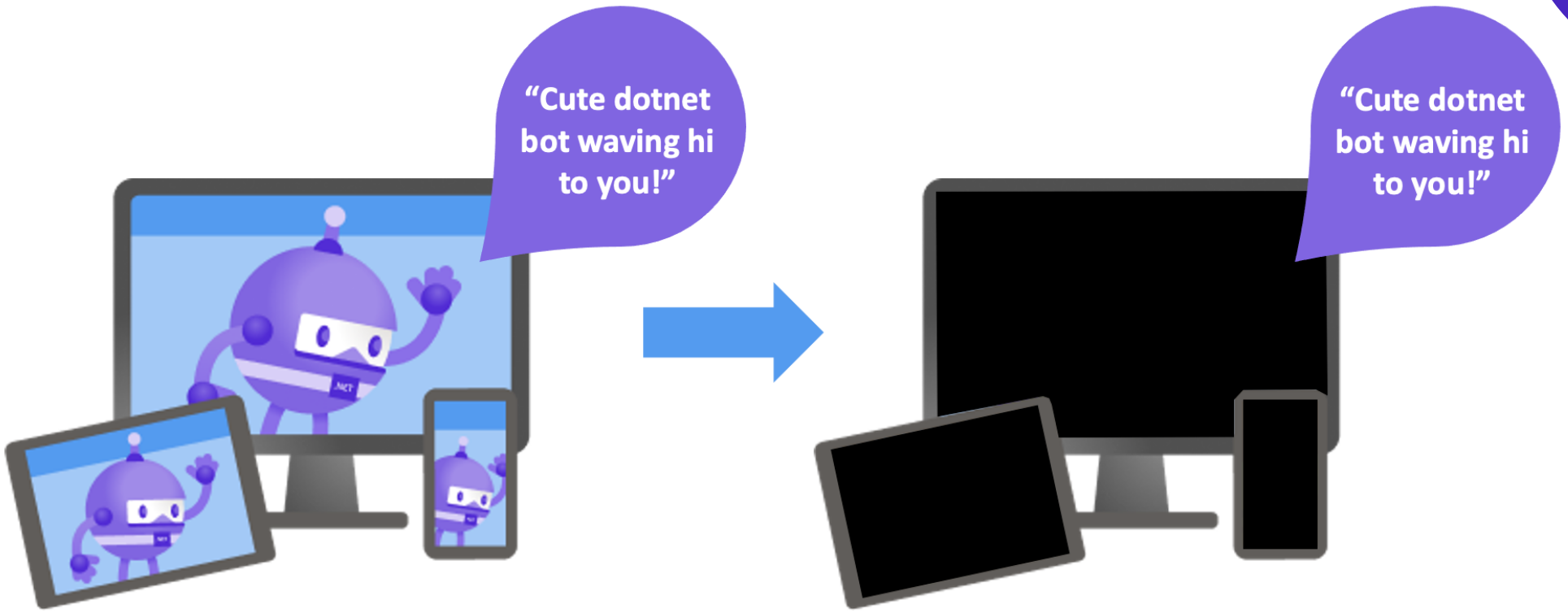
Animations



Accessibility

Semantic Properties

1



Semantic Properties




```
SemanticProperties.Description
SemanticProperties.Hint
SemanticProperties.HeadingLevel
```

```
<Image Source="logo.png"
       SemanticProperties.Description=".NET logo" />

Image image = new Image { Source = "logo.png" };
SemanticProperties.SetDescription(image, ".NET logo");
```

Semantic Extensions

2

```
SetSemanticFocus()
Announce()
```

```
Image image = new Image { Source = “logo.png” };
image.SetSemanticFocus();

SemanticScreenReader.Announce(“I am being announced!”);
```

Community Toolkit

3

SemanticEffects

SemanticOrderView

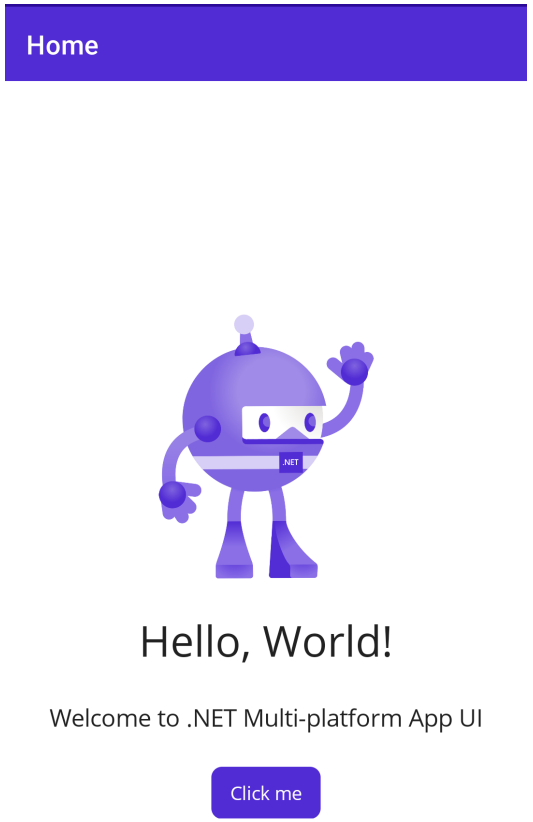
```
<xct:SemanticOrderView x:Name="sov">
    <StackLayout>
        <Label x:Name="second" Text="Second control" />
        <Button x:Name="third" Text="Third control" />
        <Label x:Name="fourth" Text="Fourth control" />
        <Button x:Name="fifth" Text="Fifth control" />
        <Button x:Name="first" Text="First control" />
    </StackLayout>
</xct:SemanticOrderView>
```

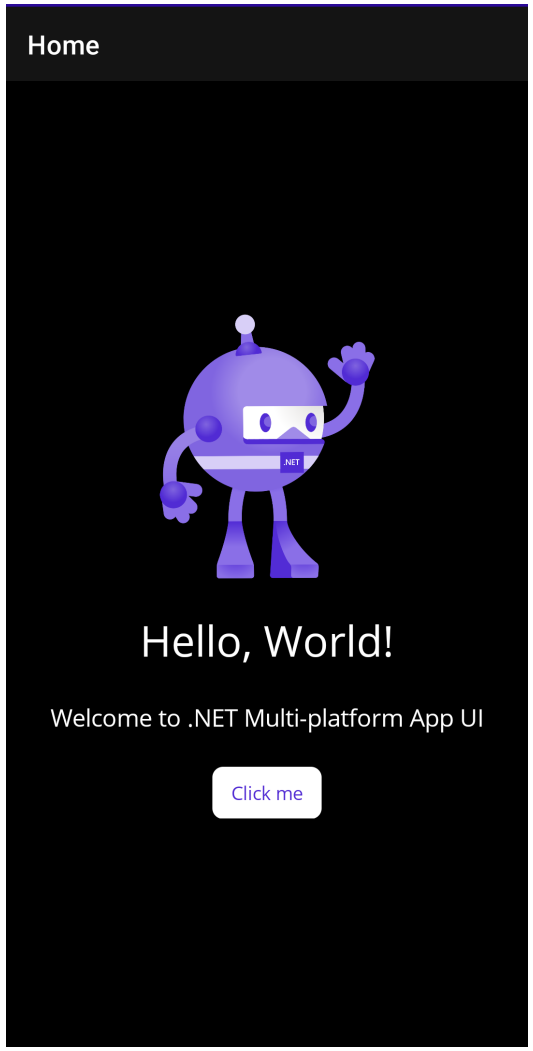
```
sov.ViewOrder = new List<View> { first, second, third, fourth, fifth };
```

https://github.com/CommunityToolkit/Maui

App Theming

4



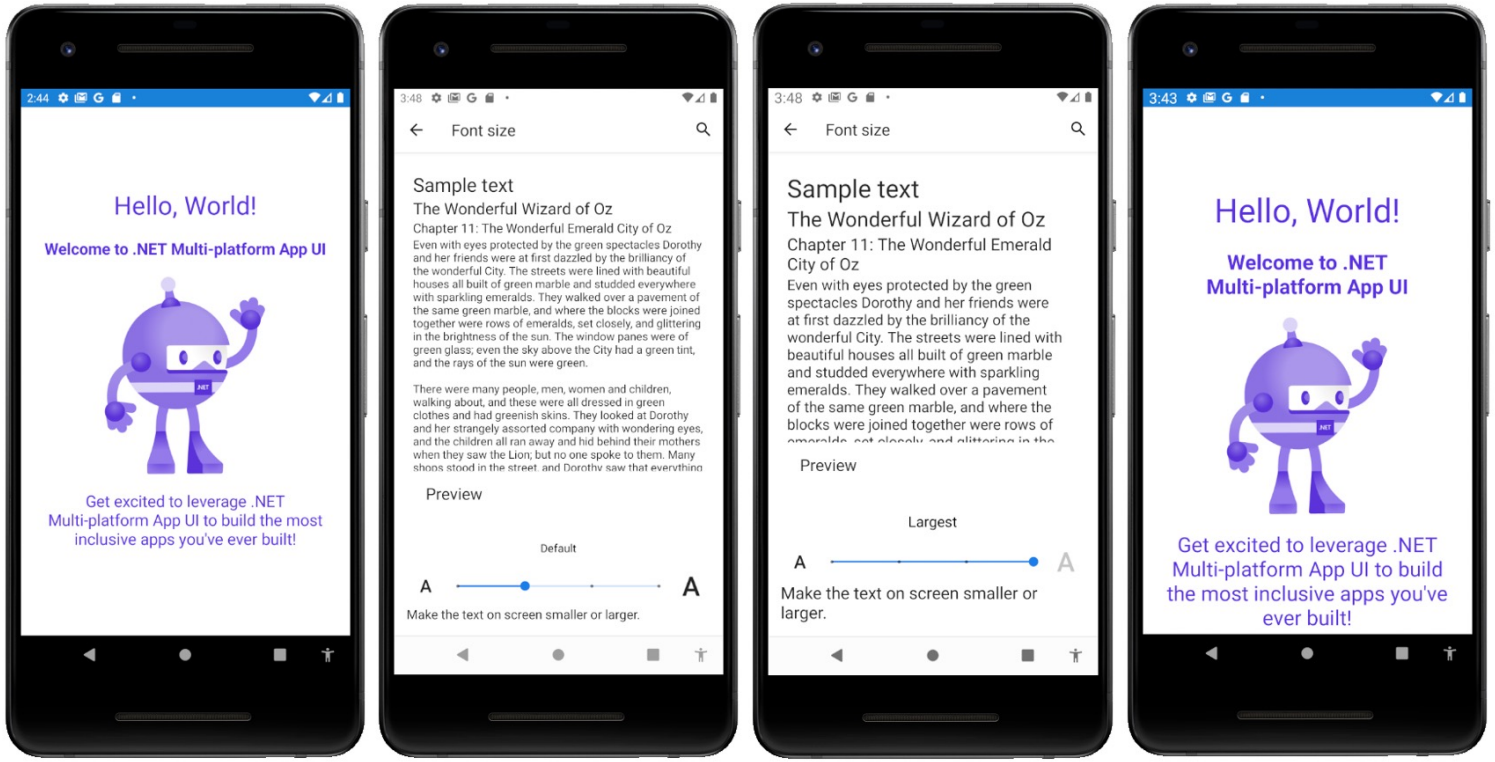



```
<Setter
    Property="BackgroundColor”
    Value="{AppThemeBinding
        Light={StaticResource White},
        Dark={StaticResource Black}}" />
```

Font Autoscaling

5

`FontAutoScalingEnabled`



Built-in Accessibility

6

Apply Accessibility traits to controls that have TapGestureRecogniz

Merged **PureWeen** merged 7 commits into `release/6.0.1xx-preview7` from `node_info_alternate` on 2 Aug 2021

Conversation 6 · Commits 7 · Checks 14 · Files changed 17



PureWeen commented on 30 Jul 2021 • edited — Member

Description of Change

If users attach a touch gesture recognizer that will activate with a single touch then we add accessibility traits that will inform readers that the element is clickable.

Add traits to iOS Checkbox so it reads as Switch #1832

Merged **PureWeen** merged 4 commits into `main` from `ios_accessible_checkbox` on 27 Jul 2021

Conversation 0 · Commits 4 · Checks 14 · Files changed 2



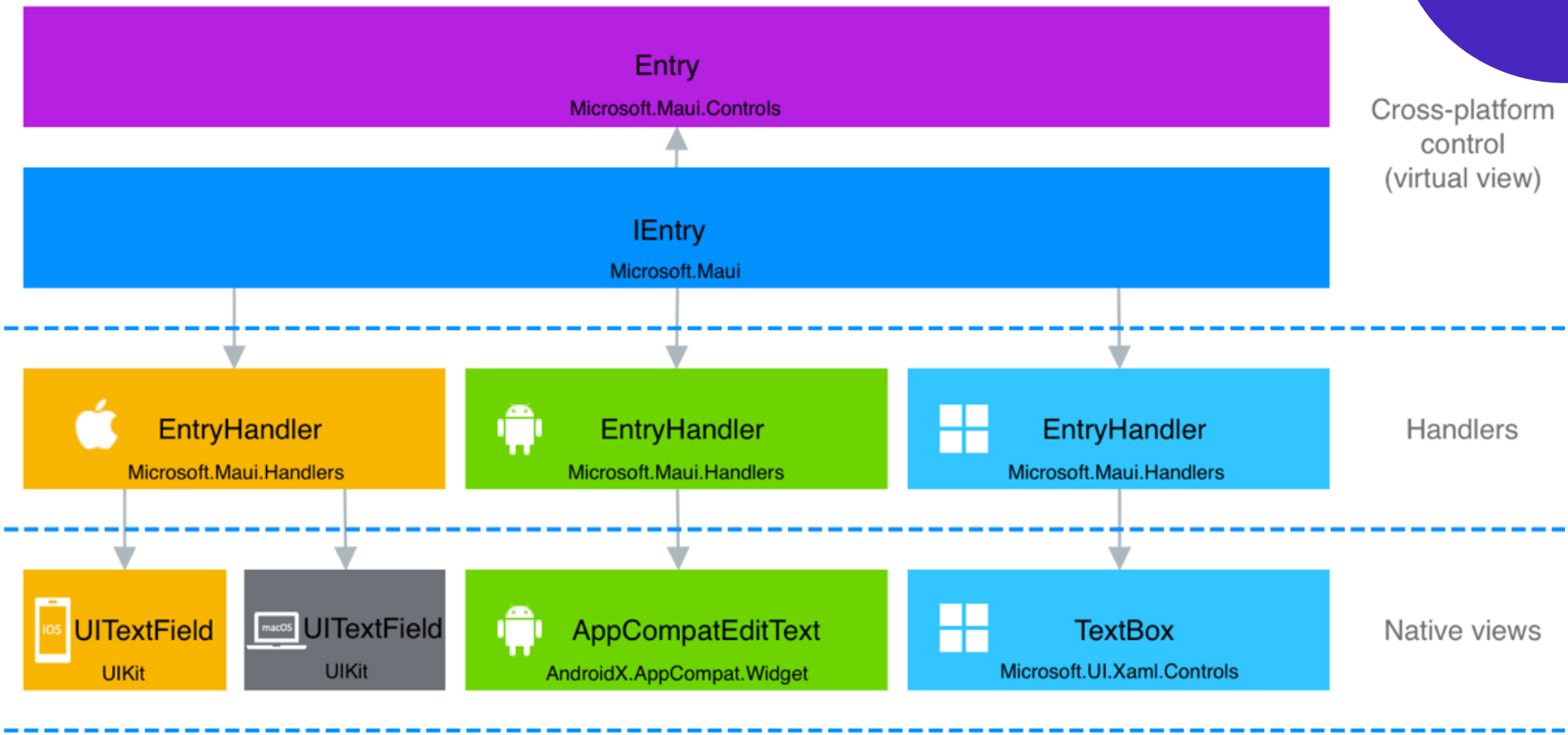
PureWeen commented on 27 Jul 2021 • edited — Member

Description of Change

iOS doesn't natively have a checkbox so to make the current checkbox more accessible we will treat it semantically like a switch

Customization via Handlers

7



Customization via Handlers



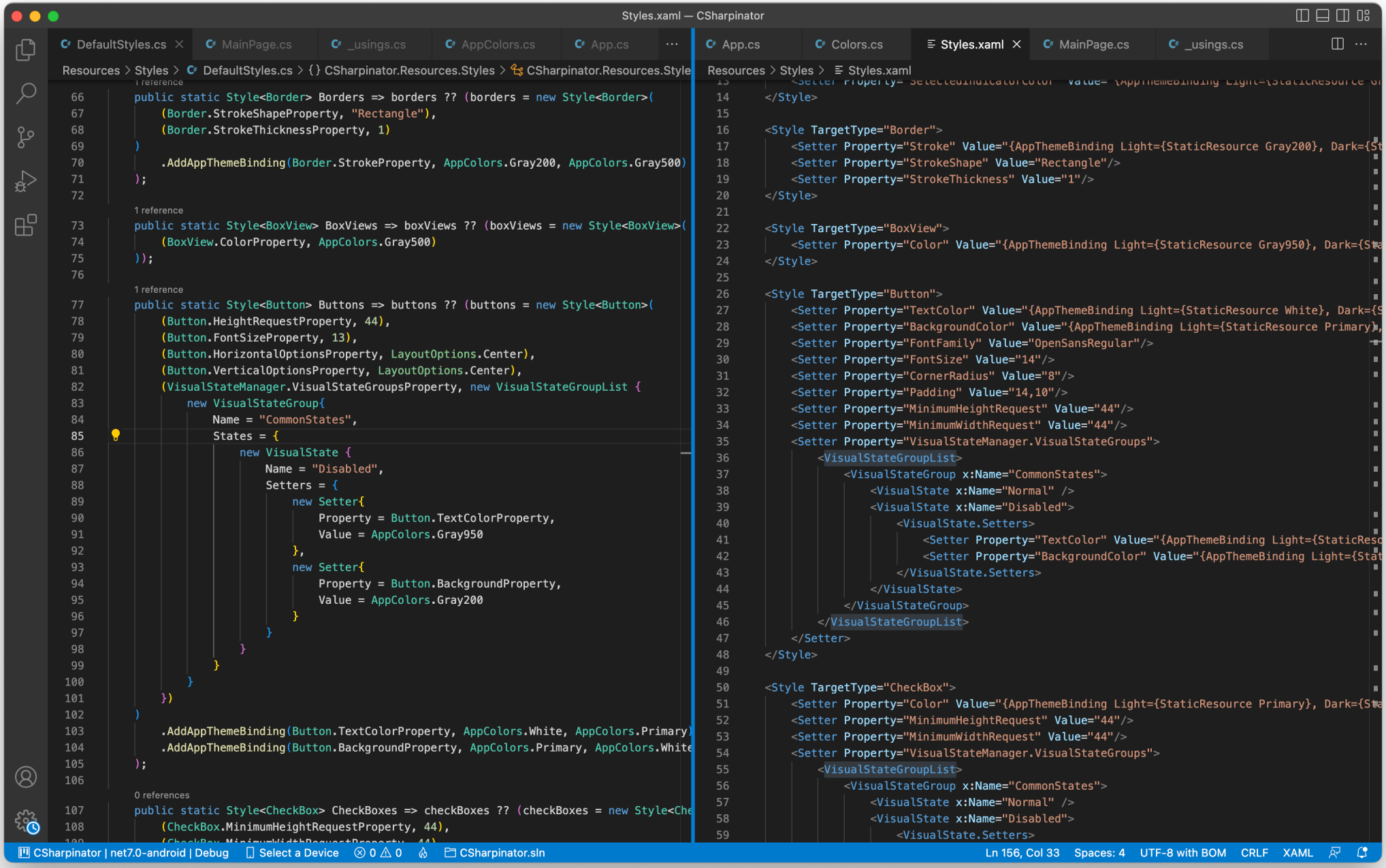

```
    Microsoft.Maui.Handlers.EntryHandler.Mapper.AppendToMapping
    (“MyEntry”, (handler, view) =>
    {
#if ANDROID
        // Android code

#elif IOS || MACCATALYST
        // iOS code

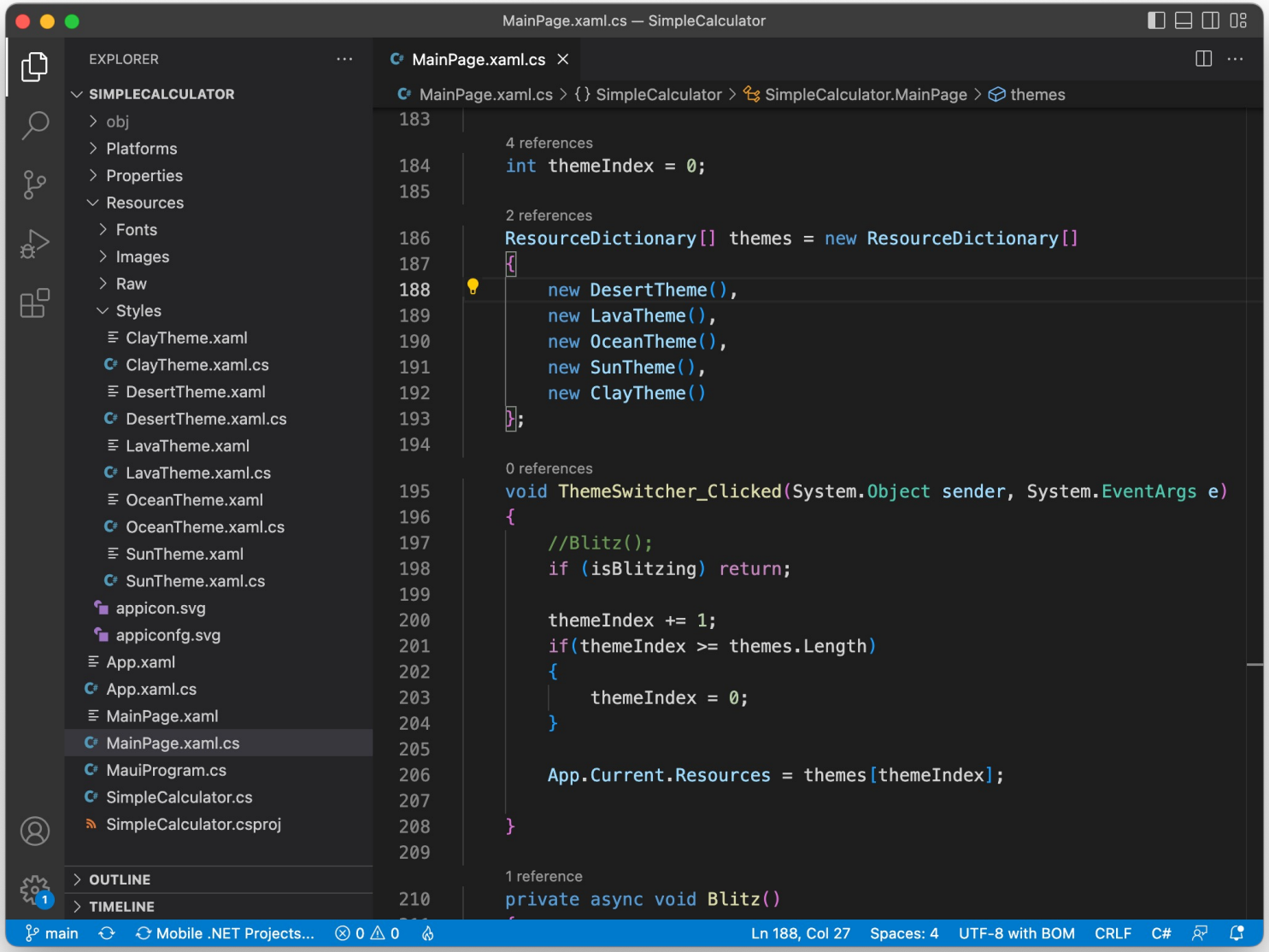
#elif WINDOWS
        // WinUI code
#endif
    });
```

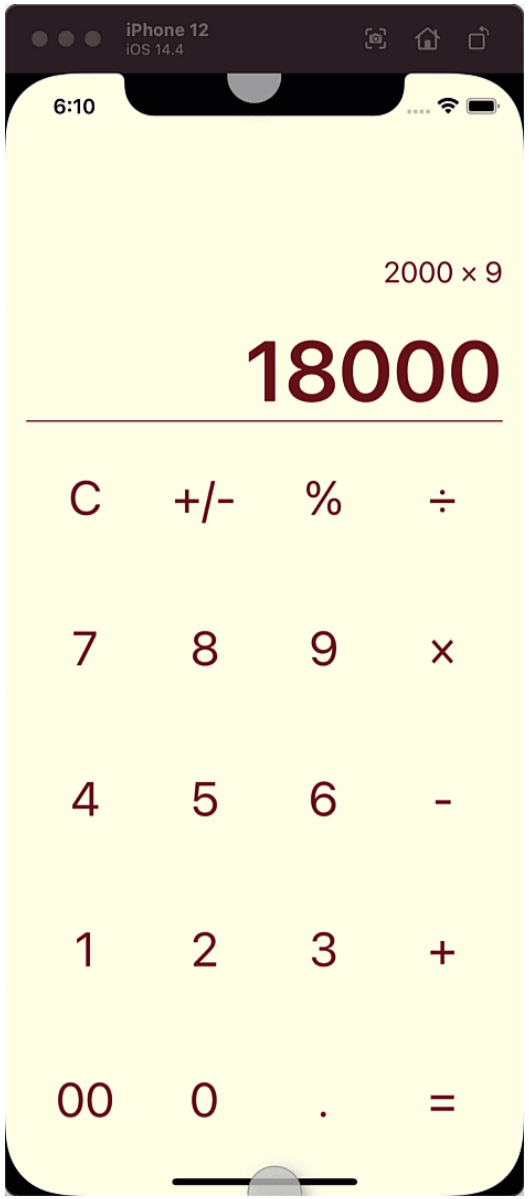
Styles and Themes

Styles


Styles.xaml — CSharpinator
DefaultStyles.cs
MainPage.cs
_usings.cs
AppColors.cs
App.cs
Colors.cs
Styles.xaml
Resources > Styles > DefaultStyles.cs > {} CSharpinator.Resources.Styles > CSharpinator.Resources.Style
Resources > Styles > Styles.xaml
CSharpinator | net7.0-android | Debug
Select a Device
CSharpinator.sln
Ln 156, Col 33
Spaces: 4
UTF-8 with BOM
CRLF
XAML

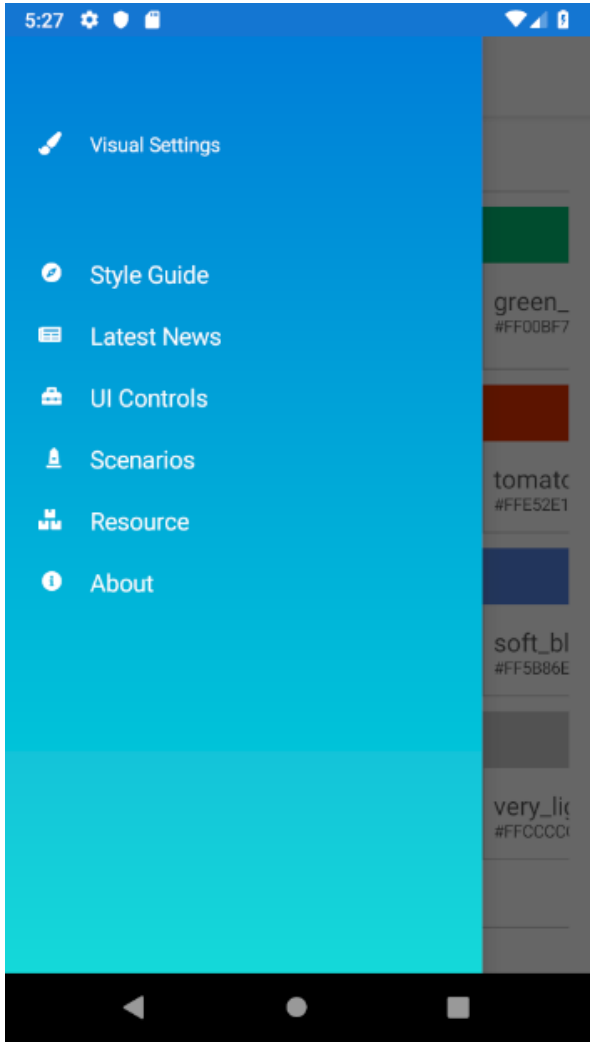
Themes

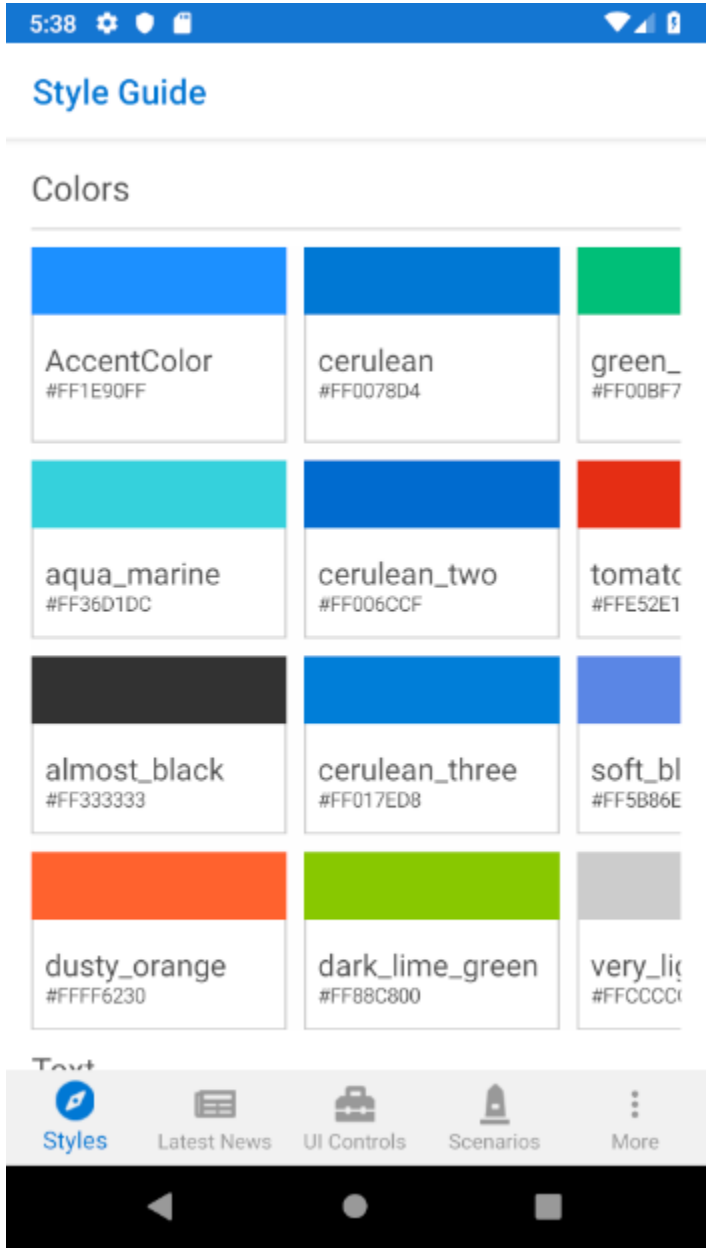

MainPage.xaml.cs — SimpleCalculator
EXPLORER
SIMPLECALCULATOR
obj
Platforms
Properties
Resources
Fonts
Images
Raw
Styles
ClayTheme.xaml
ClayTheme.xaml.cs
DesertTheme.xaml
DesertTheme.xaml.cs
LavaTheme.xaml
LavaTheme.xaml.cs
OceanTheme.xaml
OceanTheme.xaml.cs
SunTheme.xaml
SunTheme.xaml.cs
appicon.svg
appiconfg.svg
App.xaml
App.xaml.cs
MainPage.xaml
MainPage.xaml.cs
MauiProgram.cs
SimpleCalculator.cs
SimpleCalculator.csproj
OUTLINE
TIMELINE
MainPage.xaml.cs
MainPage.xaml.cs > { } SimpleCalculator > SimpleCalculator.MainPage > themes
4 references
int themeIndex = 0;
2 references
ResourceDictionary[] themes = new ResourceDictionary[]
{
new DesertTheme(),
new LavaTheme(),
new OceanTheme(),
new SunTheme(),
new ClayTheme()
};
0 references
void ThemeSwitcher_Clicked(System.Object sender, System.EventArgs e)
{
//Blitz();
if (isBlitzing) return;
themeIndex += 1;
if(themeIndex >= themes.Length)
{
themeIndex = 0;
}
App.Current.Resources = themes[themeIndex];
}
1 reference
private async void Blitz()
main
Mobile .NET Projects...
Ln 188, Col 27
Spaces: 4
UTF-8 with BOM
CRLF
C#

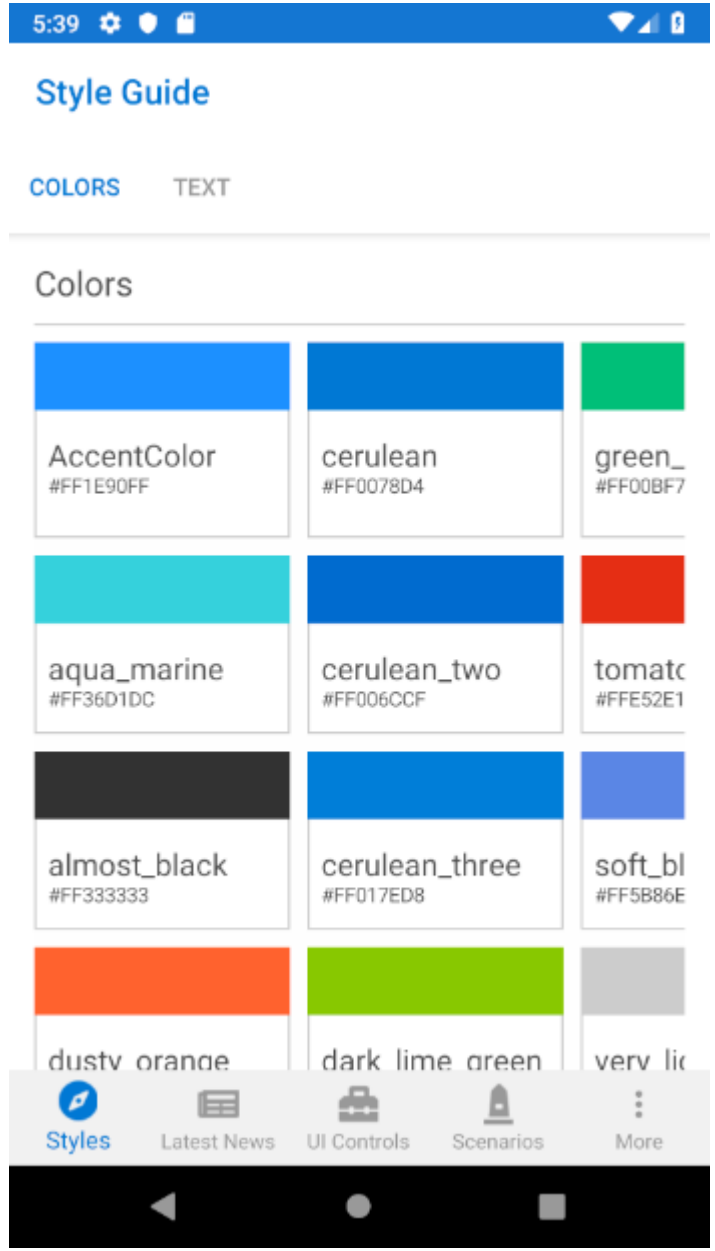

iPhone 12
iOS 14.4
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Navigation

Flyout, Tabs, and more


5:27
Visual Settings
Style Guide
Latest News
UI Controls
Scenarios
Resource
About


5:38
Style Guide
Colors
AccentColor
#FF1E90FF
cerulean
#FF0078D4
aqua_marine
#FF36D1DC
cerulean_two
#FF006CCF
almost_black
#FF333333
cerulean_three
#FF017ED8
dusty_orange
#FFFF6230
dark_lime_green
#FF88C800
Styles
Latest News
UI Controls
Scenarios
More


5:39
Style Guide
COLORS
TEXT
Colors
AccentColor
#FF1E90FF
cerulean
#FF0078D4
aqua_marine
#FF36D1DC
cerulean_two
#FF006CCF
almost_black
#FF333333
cerulean_three
#FF017ED8
Styles
Latest News
UI Controls
Scenarios
More

Shell

Tabs Made Simple

```
<TabBar>
    <Tab Title="Browse" Icon="tab_feed.png">
        <ShellContent ContentTemplate="{DataTemplate local:ItemsPage}" />
    </Tab>
    <Tab Title="About" Icon="tab_about.png">
        <ShellContent ContentTemplate="{DataTemplate local:AboutPage}" />
    </Tab>
</TabBar>
```

```
<TabbedPage x:Class="App5.Views.MainPage">

    <TabbedPage.Children>
        <NavigationPage Title="Browse">
            <NavigationPage.Icon>
                <OnPlatform x:TypeArguments="FileImageSource">
                    <On Platform="iOS" Value="tab_feed.png"/>
                </OnPlatform>
            </NavigationPage.Icon>
            <x:Arguments>
                <views:ItemsPage />
            </x:Arguments>
        </NavigationPage>

        <NavigationPage Title="About">
            <NavigationPage.Icon>
                <OnPlatform x:TypeArguments="FileImageSource">
                    <On Platform="iOS" Value="tab_about.png"/>
                </OnPlatform>
            </NavigationPage.Icon>
            <x:Arguments>
                <views:AboutPage />
            </x:Arguments>
        </NavigationPage>
    </TabbedPage.Children>

</TabbedPage>
```

Flyout Made Simple

```
<FlyoutItem Title="Browse"
    <ShellCon
</FlyoutItem>
<FlyoutItem T
    <ShellCon
</FlyoutItem>
```

```
<MasterDetailPage xmlns="http://xamarin.com/schemas/2014/
    xmlns:x="http://schemas.microsoft.com/winfx/2
    xmlns:d="http:
    xmlns:mc="http
    mc:Ignorable="
    xmlns:views="
    x:Class="App6.
  <MasterDetailPage.Maste
    <views:MenuPage />
  </MasterDetailPage.Mast
  <MasterDetailPage.Deta
    <NavigationPage>
      <NavigationPage
        <OnPlatform
          <On Pla
        </OnPlatfor
      </NavigationPag
      <x:Arguments>
        <views:Iter
      </x:Arguments>
    </NavigationPage>
  </MasterDetailPage.Deta

</MasterDetailPage>
```

```
menuItems = new List<Hom
            {
                new Home
                new Home
            };

            ListViewMenu

            ListViewMenu
            ListViewMenu
            {
                if (e.Se
                    retu

                var id =
                await Ro
            };
```

```
public partial class MainPage : MasterDetailPage
    {
        Dictionary<int, NavigationPage> MenuPages = new Dictionary<int, NavigationPage>();
        public MainPage()
        {
            InitializeComponent();

            MasterBehavior = MasterBehavior.Popover;

            MenuPages.Add((int)MenuItemType.Browse, (NavigationPage)Detail);
        }

        public async Task NavigateFromMenu(int id)
        {
            if (!MenuPages.ContainsKey(id))
            {
                switch (id)
                {
                    case (int)MenuItemType.Browse:
                        MenuPages.Add(id, new NavigationPage(new ItemsPage()));
                        break;
                    case (int)MenuItemType.About:
                        MenuPages.Add(id, new NavigationPage(new AboutPage()));
                        break;
                }
            }

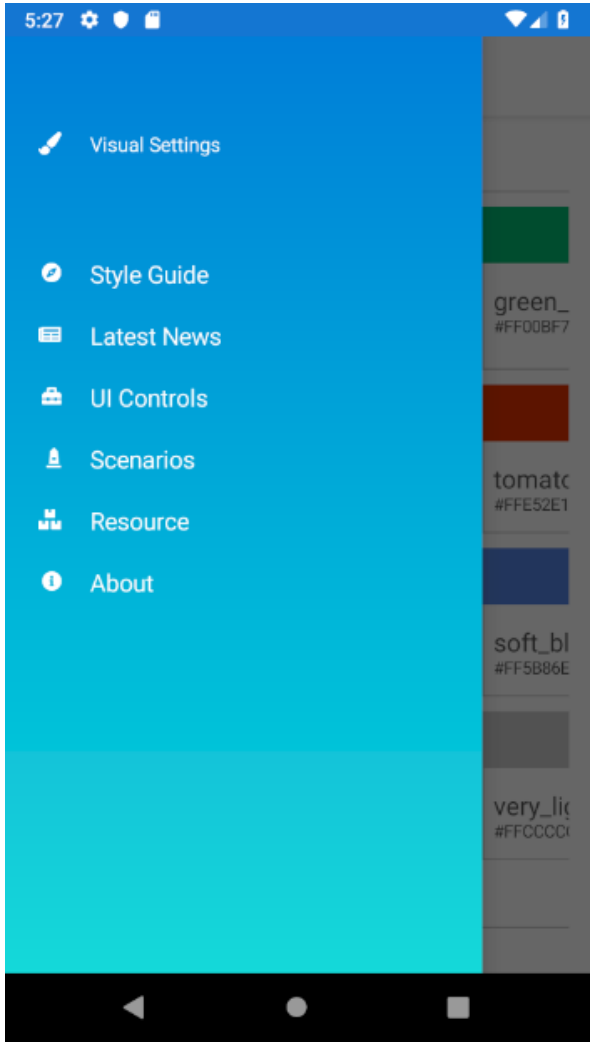
            var newPage = MenuPages[id];

            if (newPage != null && Detail != newPage)
            {
                Detail = newPage;

                if (Device.RuntimePlatform == Device.Android)
                    await Task.Delay(100);

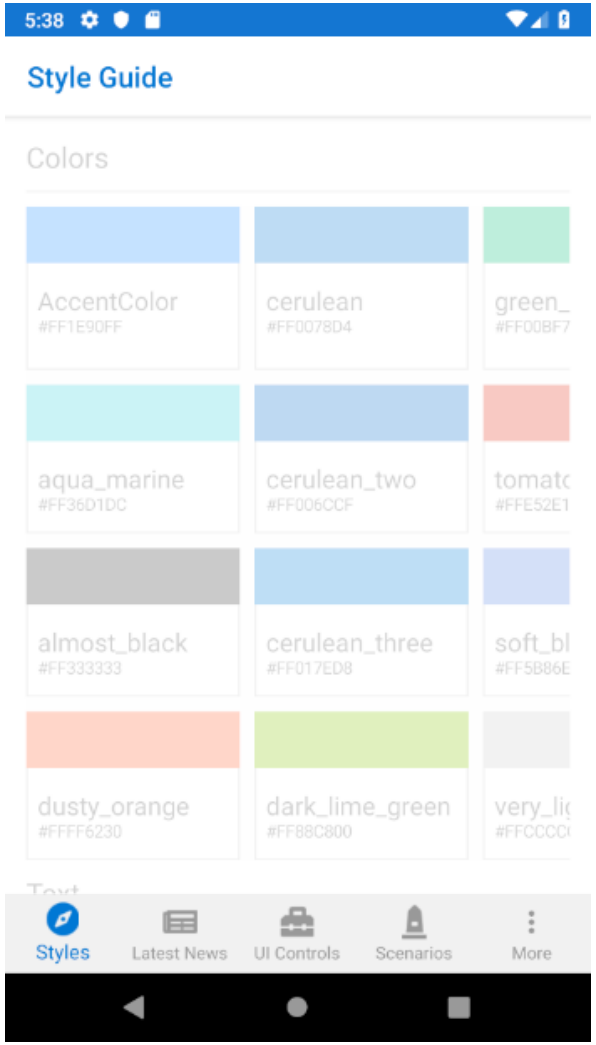
                IsPresented = false;
            }
        }
    }
```

```
</ContentPage>
```



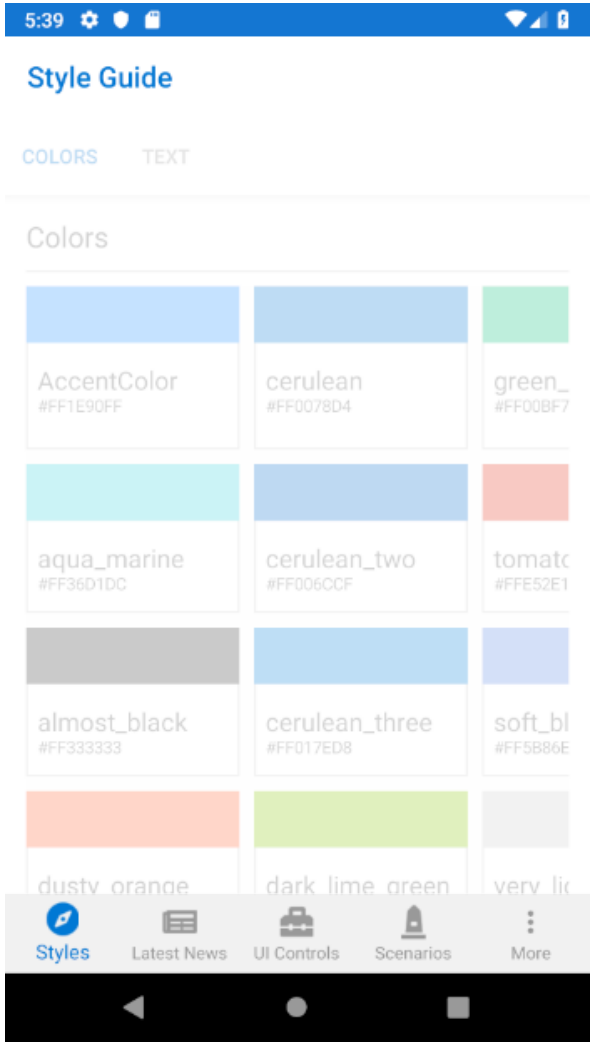

```
<Shell>
    <FlyoutItem
        Title="Style Guide"
        Icon="Compass.png">
        <ShellContent
            ContentTemplate="{DataTemplate p:StyleGuidePage}"/>
    </FlyoutItem>
</Shell>
```

Shell




```
<Shell>
    <TabBar>
        <Tab
            Title="Styles"
            Icon="Compass.png">
            <ShellContent
                ContentTemplate="{DataTemplate
                                  p:StyleGuidePage}"/>
        </Tab>
    </TabBar>
</Shell>
```

Shell



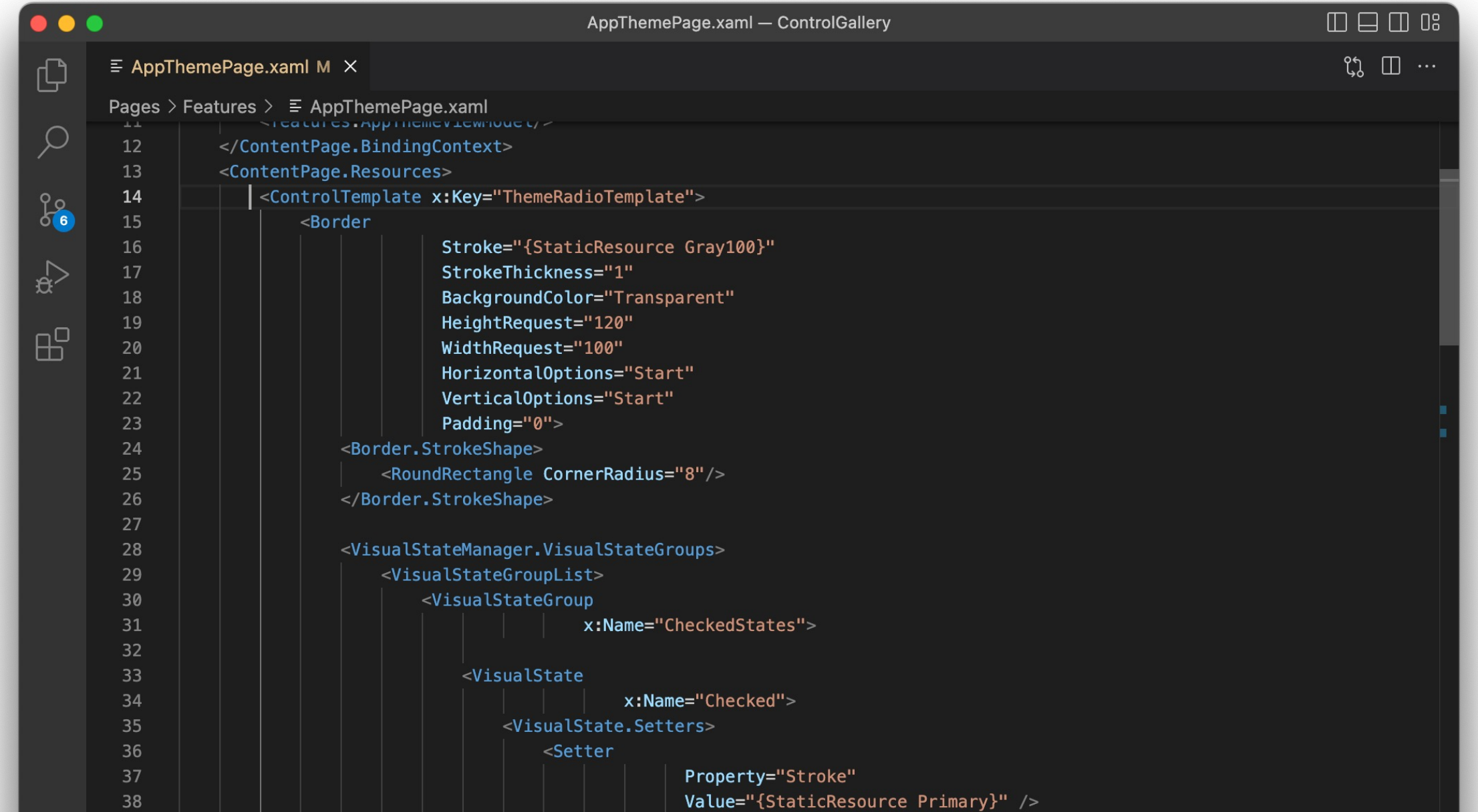

```
<Shell>
    <TabBar>
        <Tab
            Title="Style Guide"
            Icon="Compass.png">
            <ShellContent
                Title="Colors"
                ContentTemplate="{DataTemplate p:ColorsPage}"/>
            <ShellContent
                Title="Text"
                ContentTemplate="{DataTemplate p:TextPage}"/>
        </Tab>
    </TabBar>
</Shell>
```

Shell

Control Customizations

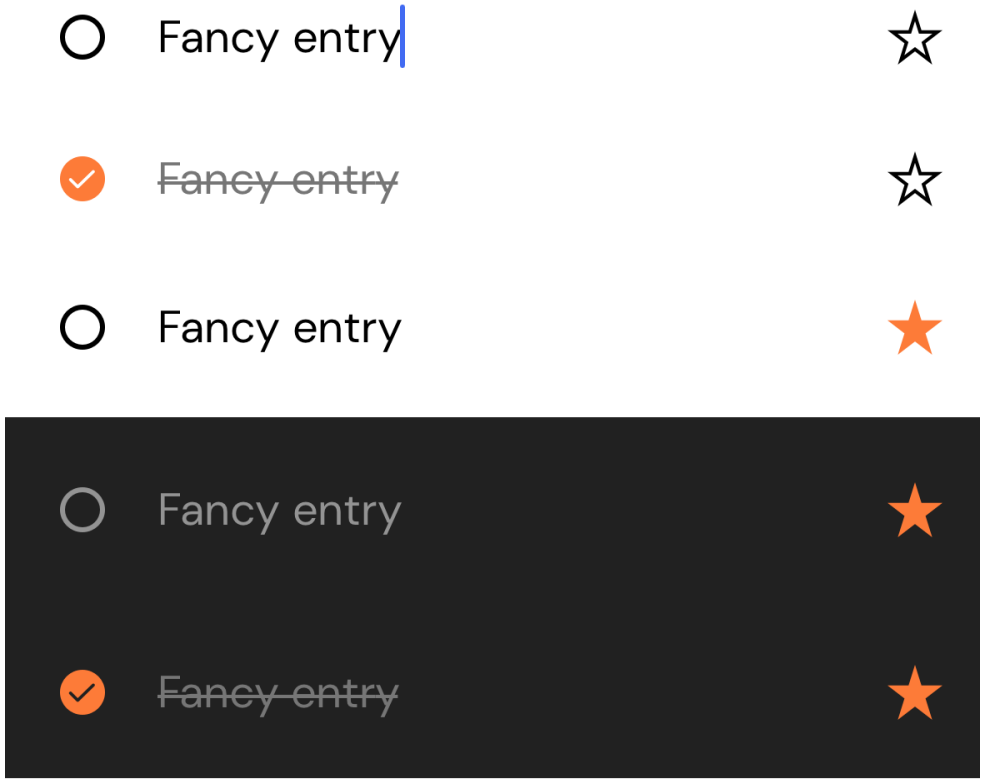
You have all the control

Control Templating



Control Composition

Grid
- Checkbox
- Entry
- Label
- ImageButton



https://github.com/davidortinau/ToDoEntry/

Handlers

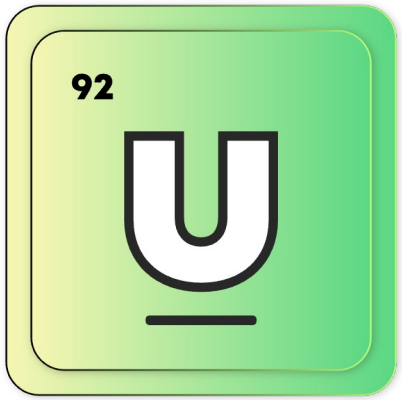
- **PrependToMapping**, which modifies the mapper for a handler before the .NET MAUI control mappings have been applied.
- **ModifyMapping**, which modifies an existing mapping.
- **AppendToMapping**, which modifies the mapper for a handler after the .NET MAUI control mappings have been applied.

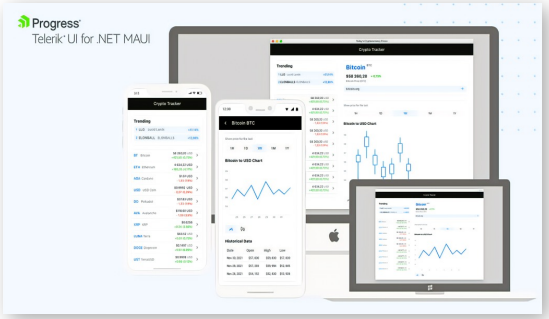
https://learn.microsoft.com/en-us/dotnet/maui/user-interface/handlers/customize

Other Options

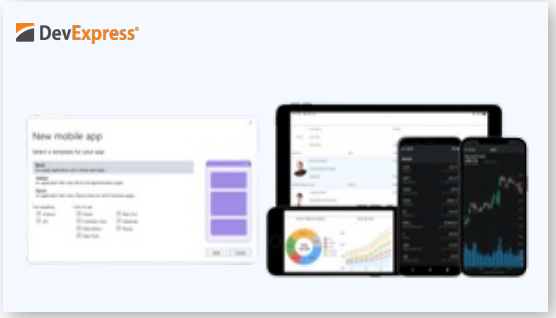
- **Draw** your own UI with Microsoft.Maui.Graphics
- **Choose** from any of the awesome control libraries in the .NET ecosystem.









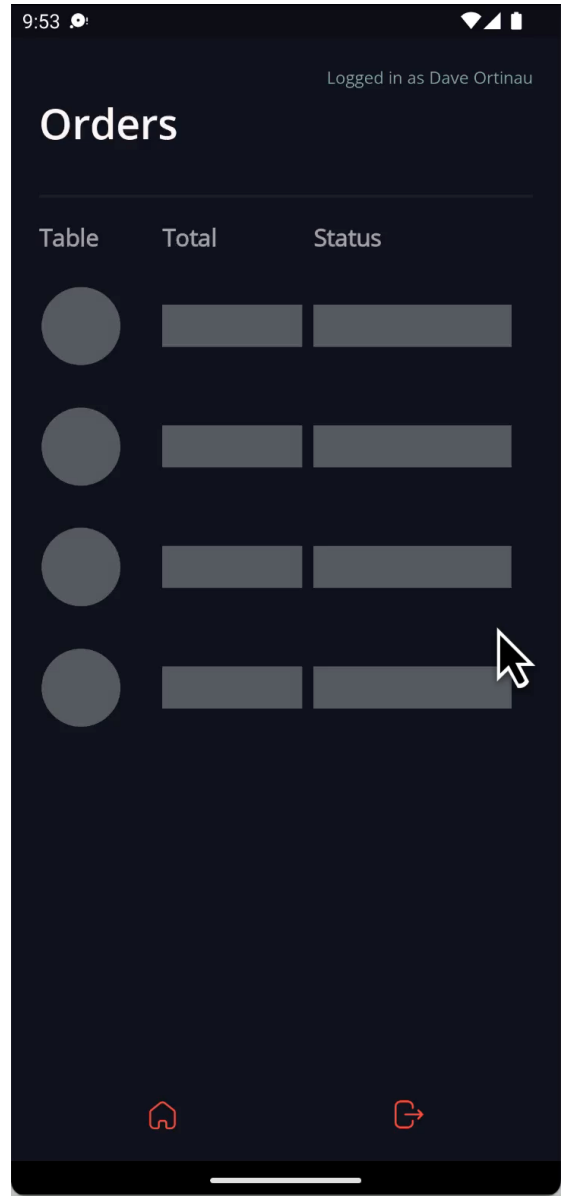




State and Animation

StateContainer

- Show placeholders while you are loading content
- More meaningful than indeterminant spinners
- BoxView is your friend
- Part of .NET MAUI Community Toolkit
- Animations courtesy of AlohaKit.Animations




Show me a demo!

Resources

- https://snppts.dev
- https://github.com/jsuarezruiz/awesome-dotnet-maui
- https://github.com/dotnet/maui-samples
- https://github.com/davidortinau/

Thanks! Let’s talk.

david.ortinau@microsoft.com

@davidortinau



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